Defense Script — Short Lines to Memorize

Two lines max per slide. Simple words. Speak slow.

Slide 1 — Title

Good morning everyone. Today we will present our work on comparing imputation and clustering pipelines on incomplete newspaper suicide data from Bangladesh.

Slide 2 — The Team

We are a team of five from the CSE department of BUBT. Our names and IDs are shown on the screen.

Slide 3 — Overview

Our presentation has six sections. Background, literature, research question, methodology, results, and findings with limitations.

Slide 4 — Section: Background

Let me start with the background.

Slide 5 — Suicide in Bangladesh

Suicide in Bangladesh is a serious public-health crisis. But the country has no national surveillance system to track it.

Slide 6 — The numbers

Bangladesh Police recorded 73,597 suicide deaths between 2020 and 2024. That is about 14,719 deaths per year, or 40 deaths every single day.

Slide 7 — The true figure is higher

But the real number is even higher than this. Because of stigma, cultural barriers, and no surveillance system, many cases are never reported.

Slide 8 — Section: The Data Problem

Now let me explain the data problem we faced.

Slide 9 — Newspapers are the only source

In Bangladesh, newspapers are the only public source of suicide case data. There is no hospital or government dataset available.

Slide 10 — 50 to 90 percent missing

But newspaper data is broken. Around 50 to 90 percent of critical fields are missing. Fields like age, occupation, mental health, location, and weather are often empty.

Slide 11 — Standard ML cannot run

Because of this, standard machine learning cannot run on this data. Dropping rows wastes most of the dataset, and simple mean filling creates bias.

Slide 12 — Section: Literature Review

Now I will discuss the literature review and the research gap.

Slide 13 — Bangladeshi research

Two important studies by Arafat and team looked at Bangladeshi newspaper data. But they only did manual grouping or aggregate statistics. No algorithm was used.

Slide 14 — Global research

Globally, Xiao and Wong applied machine learning and clustering on suicide data. But their work was on the US and Hong Kong, not on Bangladesh.

Slide 15 — The Research Gap

We found five gaps in the literature. No clustering on Bangladeshi news data, no multi-method imputation, only single algorithm used, weak metrics, and no comparison of predictor strategies.

Slide 16 — Section: Research Framing

This brings us to our research framing.

Slide 17 — Research Question

Our research question is simple. Which combination of missing-data handling and unsupervised learning gives the most stable and interpretable cluster structure from this data?

Slide 18 — Objectives

We had four objectives. Build the dataset, benchmark 32 pipelines, design a balance-aware score, and find the best pipeline with honest caveats.

Slide 19 — Section: Methodology

Now let me walk you through the methodology.

Slide 20 — End-to-end pipeline

This figure shows our full pipeline at a high level. It goes from data collection to preprocessing, encoding, imputation, clustering, and evaluation.

Slide 21 — Dataset

We built a dataset of 1,617 records covering 2017 to 2025. 760 came from a public Kaggle dataset and 857 we collected manually.

Slide 22 — New attributes

We added nine new attributes to enrich the data. These cover family, profession, financial condition, and mental-health history.

Slide 23 — External API enrichment

We used two APIs to fill in missing context. Geopy gave us geocoding, and Meteostat gave us weather data.

Slide 24 — Encoding strategy

We used different encoding for different feature types. Frequency for ordinal, one-hot for low-cardinality, and label encoding for MissForest.

Slide 25 — Imputation workflow

This diagram shows the full imputation workflow. We used three methods and two predictor strategies, giving six imputed datasets.

Slide 26 — Three imputation methods

The three methods are MICE, KNN, and MissForest. Each one uses a different statistical approach to fill missing values.

Slide 27 — Two predictor strategies

We used two predictor strategies. Context-aware uses only correlated columns, and no-context uses every available column.

Slide 28 — Clustering workflow

This diagram shows the clustering workflow. Every dataset goes through scaling, PCA, grid search, and final scoring.

Slide 29 — Four clustering algorithms

We used four clustering algorithms from four different paradigms. K-Means for centroid, Agglomerative for hierarchical, DBSCAN for density, and Spectral for graph.

Slide 30 — Hyperparameter ranges

For each algorithm we tested a wide range of hyperparameters. This made the comparison fair and reproducible.

Slide 31 — Standard scores fail

Standard scores like Silhouette can reward bad solutions. For example, DBSCAN can label 99 percent of cases as noise and still score high.

Slide 32 — Six-component composite score

To fix this we built a six-component composite score. The balance penalty has the biggest weight, so degenerate clusters are demoted.

Slide 33 — Here is what we built

And this is the full detailed pipeline showing everything from start to finish. Eight datasets times four algorithms equals 32 clustering pipelines.

Slide 34 — Section: Results

Now let me share the results.

Slide 35 — All 32 pipelines ranked

We ranked all 32 pipelines by composite score. Agglomerative with MissForest no-context came first, with a score of 27.30 out of 32.

Slide 36 — Composite heatmap

This heatmap shows how every pipeline performed on every metric. Darker red means higher score.

Slide 37 — Group averages

On average, MissForest imputation beat MICE and KNN. And Spectral and K-Means did better than Agglomerative overall.

Slide 38 — The top pipeline

The winning pipeline is Agglomerative with MissForest no-context. It got 27.30 out of 32 and made two clear clusters.

Slide 39 — Two clusters

The two clusters had 759 and 858 records. This is a very balanced split.

Slide 40 — Geographic distribution

This map shows where each cluster is located in Bangladesh. Cluster 0 spreads across the country, while cluster 1 is more concentrated in big cities.

Slide 41 — Cluster profiling

Cluster 0 is mostly semi-urban with hometowns like Bogura and Rajshahi. Cluster 1 is more rural but with hometowns in Dhaka and Chattogram.

Slide 42 — Reason and age comparison

The two clusters also differ in reason and age distribution. Cluster 0 is wider in age, while cluster 1 peaks at younger adults.

Slide 43 — Section: Honest Findings

Now I want to share an honest finding from our work.

Slide 44 — An honest finding

The two clusters actually match the data-collection boundary. Not two real epidemiological groups.

Slide 45 — How the cohorts map

Cluster 0 is the 2020 Kaggle subset, and cluster 1 is our manually collected data. But the algorithm never saw the year. So this split is emergent, and it works as an internal consistency check.

Slide 46 — Section: Ethical Concern and Timeline

Next, I will cover our ethical safeguards and project timeline.

Slide 47 — Ethical safeguards

We took several steps to protect privacy. Names removed, addresses coarsened, URLs dropped, and methods stored only in encoded form.

Slide 48 — Project timeline

The full project took 25 weeks. You can see how each phase was done step by step.

Slide 49 — Section: Closing

And now the closing part.

Slide 50 — Limitations

We have some honest limitations. Newspaper bias, small dataset, no ground truth, sparse weather data, and limited compute.

Slide 51 — Future work

In future we want to add SHAP analysis, expand the data, and try causal modelling. BanglaBERT for free text and a real-time alerting system are also next steps.

Slide 52 — Contribution

Our main contributions are four. The largest curated Bangladeshi suicide dataset, a reproducible benchmark, a balance-aware metric, and full open-source release.

Slide 53 — Repository

You can find all our code, data, and notebooks on GitHub. The QR code on the screen takes you to the repository.

Slide 54 — Thank You

Thank you so much for listening. We are happy to take your questions now.